

SHAMIK DEY

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LinkedIn | GitHub | Google Scholar

Professional Experience

Research Assistant

Dec 2025 - Present

Ahsanullah University of Science and Technology (AUST)

AUST Internal Research Grant (AIRG): Publicly Accessible Ripe-Unripe Crop Image Dataset & AI-Based Crop Maturity Detection System

Leading end-to-end design of a maturity-classification pipeline tailored to Bangladeshi agricultural crops, from data acquisition and multi-class annotation protocol design through CNN/object-detection model development and evaluation. Structuring the dataset for public release to support reproducible agri-AI research.

Software Engineer

Jun 2025 - Present

Sincos Automation Technologies Ltd.

Designed and deployed a voltage anomaly detection and time-series forecasting pipeline for an industrial monitoring application, covering preprocessing, model development, and real-time inference deployment. Concurrently building the frontend architecture for an OEE analytics platform (Angular, Tailwind CSS).

Jr. Software Engineer

Jan 2024 - Jun 2025

Sincos Automation Technologies Ltd.

Contributed to OEE software design and a voltage anomaly detection system, gaining early exposure to applied ML deployment alongside full-stack development.

Publications

1. **Shamik Dey**, Adnan Quaium, Jeba Maliha, Md Mehedi Hasan, Md. Abdullah-Al-Wadud, Saifur Rahman Sabuj, and Jia Uddin, “**ResShuffle-ECAFusion: ECA-Guided Dual-Backbone Feature Fusion for Explainable and Uncertainty-Aware Differentiation of Glioma and Meningioma from Brain MRI**”, *Scientific Reports Journal (Under Review)*
2. **Shamik Dey**, Adnan Quaium, Saiful Islam, Abdus Shakur Siam, “**EyeDiseaseNet: An Explainable and Resource-Efficient Self-Attention Hybrid Network for Multi-Class Ocular Disease Classification**”, *BMC Ophthalmology Journal (Under Review)*
3. **Shamik Dey**, Ayon Rahman, Sadique Billah, Adnan Quaium, “**Solar Irradiance Forecasting with a Compact Gradient Boosting Ensemble**”, 2025 28th International Conference on Computer and Information Technology (ICCIT) DOI: [10.1109/ICCIT68739.2025.11490233](https://doi.org/10.1109/ICCIT68739.2025.11490233)
4. **Shamik Dey**, Adnan Quaium, “**Leakage-Safe Multi-View Graph Construction for Robust Intrusion Detection in Edge-IIoT Networks**,” 12th International Conference on Networking, Systems, and Security (NSysS '25). ACM, New York, NY, USA DOI: [10.1145/3777555.3777572](https://doi.org/10.1145/3777555.3777572)
5. Md Ehsanul Haque, Abrar Fahim, **Shamik Dey**, SA Jahan, SM Islam, Sakib Rokoni, MS Morshed, “**A Modified VGG19-Based Framework for Accurate and Interpretable Real-Time Bone Fracture Detection**,” 16th International IEEE Conference on Computing, Communication and Networking Technologies (ICCCNT) DOI: [10.48550/arXiv.2508.03739](https://doi.org/10.48550/arXiv.2508.03739)
6. Abrar Fahim, **Shamik Dey**, Md Nurul Absur, Md Kamrul Siam, Md Tahmidul Huque, Jafreen Jafor Godhuli, “**Optimized Approaches to Malware Detection: A Study of Machine Learning**

and Deep Learning Techniques,” 2025 IEEE 14th International Conference on Communication Systems and Network Technologies (CSNT) DOI: [10.1109/CSNT64827.2025.10968061](https://doi.org/10.1109/CSNT64827.2025.10968061)

7. SM Mozumder, Tabassum Alamgir Nishat, Bayezid Hasan Bhuiyan, Md Rahabbul Khan, **Shamik Dey**, Adnan Quaium. “**Improving Urban Mobility in Dhaka: Machine Learning-Based Traffic Prediction**,” 2025 International Conference on Electrical, Computer and Communication Engineering (ECCE) DOI: [10.1109/ECCE64574.2025.11013936](https://doi.org/10.1109/ECCE64574.2025.11013936)
8. FM Mahmudul Haque, Humayra Tabassum, Md Fazal Amin, Md Nazrul Islam Jony, **Shamik Dey**, Adnan Quaium “**Short Paper: Predicting and Analyzing EV Energy Consumption in Bangladesh: A Machine Learning Approach**,” 11th International Conference on Networking, Systems, and Security (NSysS '24). ACM, New York, NY, USA DOI: [10.1145/3704522.3704553](https://doi.org/10.1145/3704522.3704553)

Projects

Voltage Anomaly Detection and Prediction System

- Built an end-to-end machine learning pipeline for voltage anomaly detection and time-series forecasting, including data preprocessing, feature engineering, model development, and evaluation.
- Deployed the predictive model for real-time inference, providing automated anomaly alerts and voltage trend prediction for proactive system monitoring.

Production-Ready Multi-Agent AI System with FastAPI

- Engineered a scalable multi-agent AI platform using FastAPI, featuring Retriever, Summarizer, and Orchestrator agents for intelligent information retrieval and response generation.
- Implemented REST APIs, agent orchestration, automated testing (PyTest), and modular deployment using Conda with comprehensive developer documentation.

Adaptive Multi-Agent Mathematics Tutoring System

- Developed an intelligent multi-agent tutoring system that automatically routes mathematical queries to specialized reasoning agents based on educational level.
- Designed modular routing architecture, context-aware reasoning pipelines, and agent-specific explanation strategies for personalized learning experiences.

DayMate: LangGraph-Based AI Personal Planning Assistant

- Developed a LangGraph-powered AI assistant integrating weather services, news retrieval, conversational memory, and reasoning workflows for personalized daily planning.
- Leveraged tool-calling agents, ChromaDB memory, external APIs, and web scraping to generate context-aware recommendations based on user preferences and real-time events.

Education

B.Sc. in Electrical and Electronic Engineering

July 2019 - November 2023

Ahsanullah University of Science and Technology

CGPA: 3.293/4.00

Technical Skills & Strengths

Programming Languages:	Python, JavaScript.
Web Technologies:	Angular, React JS, FastAPI, Tailwind CSS, BootStrap, WordPress, PostgreSQL, MongoDB.
Libraries & Frameworks:	Pytorch, TensorFlow, OpenCV, LangChain.
Design & Verification Tools:	UI/UX Design, Cadence Virtuoso, Innovus, Quartus, Proteus, AutoCAD, PSpice, Arduino.
Cloud & General Tools:	AWS, Docker, Git, Google-Suite.

Leadership & Volunteering

Undergraduate Thesis Instructor

Feb 2023 - Present

Ahsanullah University of Science and Technology (AUST)

Appointed by Assistant Professor Md. Adnan Quaium to mentor undergraduate final-year students. Providing technical guidance on Machine Learning workflows, dataset selection, and experimental setup. Assisting students in refining their thesis writing and preparing manuscripts for peer-reviewed publication.

Technical Operations, Management & Session Anchor

Dec 2023

26th International Conference on Computer and Information Technology (ICCIT)

Treasurer

Feb 2023 - Jan 2024

IEEE AUST Student Branch

Assistant Webmaster

Mar 2022 - Feb 2023

IEEE AUST Student Branch

Significant Courses

Major Courses in B.Sc.:

Computer Architecture, VLSI - I, VLSI - II, Processing and Fabrication Technology, Computer Architecture, Power Electronics, Multimedia Communication.

Professional Certifications:

Machine Learning Specialization (Stanford / DeepLearning.AI), **Deep Learning Specialization** (DeepLearning.AI), **AI for Medical Diagnosis** (DeepLearning.AI), **Advanced Computer Vision with TensorFlow** (IBM).

Awards

Champion: Undergraduate Thesis Paper Contest at International Student LED Conference

International Student LED Conference 2023 is a co-located conference of IEEE ICCIT 2023 and IEEE ICPS 2023.

Reference

Md. Adnan Quaium

Assistant Professor, Department of Electrical and Electronic Engineering

Ahsanullah University of Science and Technology, Dhaka, Bangladesh

Email: adnan.eee@aust.edu